

Supplementary Information: Validated Synthetic Patient Generation for Small Longitudinal Cohorts

Supplementary Figures

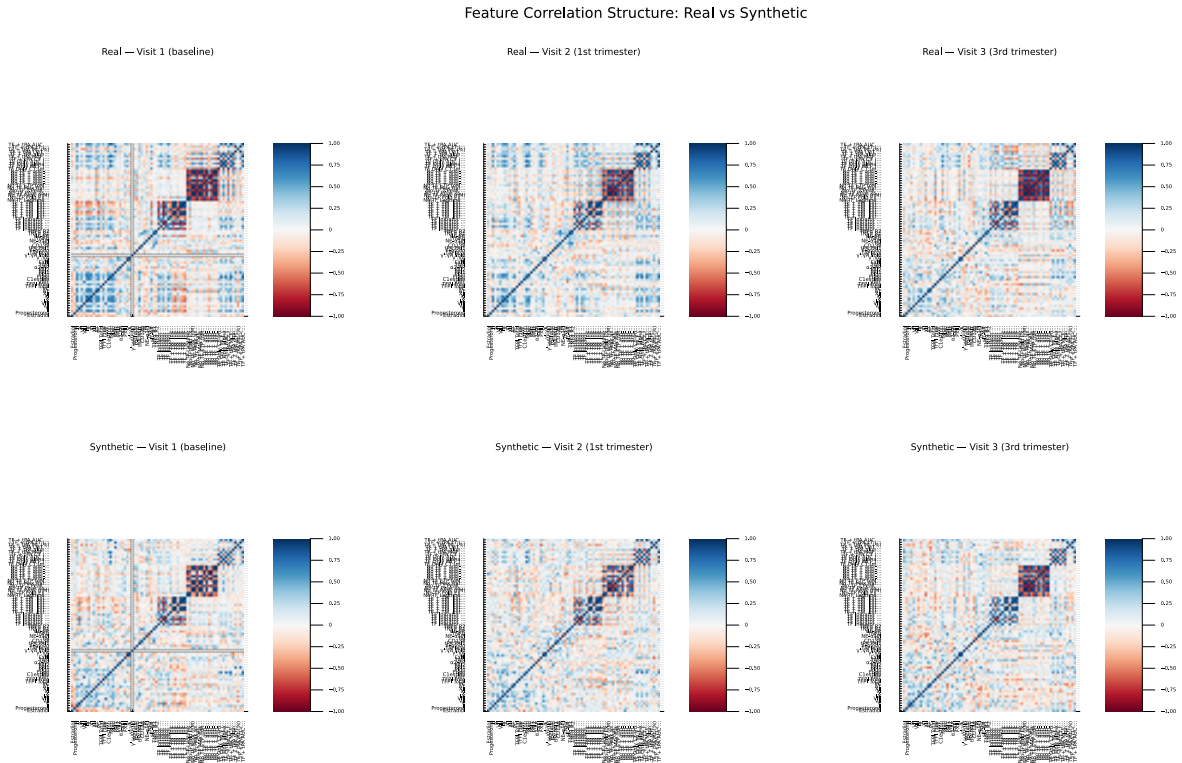


Figure S1: **Within-visit correlation heatmaps.** Pearson correlation matrices for real and SA-generated synthetic patients within each visit (72 features per visit). SA preserves the within-visit correlation structure across all three visits.

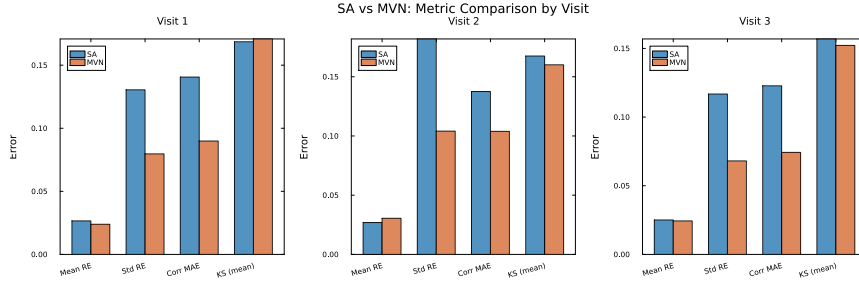


Figure S2: **SA vs. MVN quantitative comparison.** Per-visit comparison of mean relative error, standard deviation relative error, KS statistic, and correlation MAE between SA and MVN generation methods. Both methods perform comparably on marginal statistics (Level 1), but diverge on structural measures (correlation MAE).

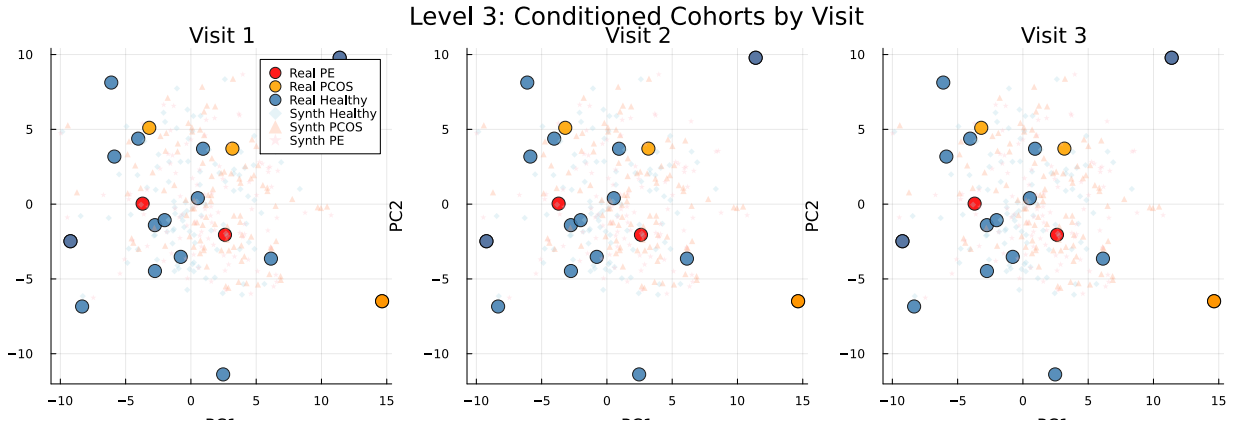


Figure S3: **Conditioned generation by visit.** PCA projections of condition-specific synthetic cohorts (Healthy, PCOS, PE) separated by visit (V1, V2, V3). The condition-specific clustering observed in the pooled projection (Fig. 5 in main text) is maintained within each individual visit.

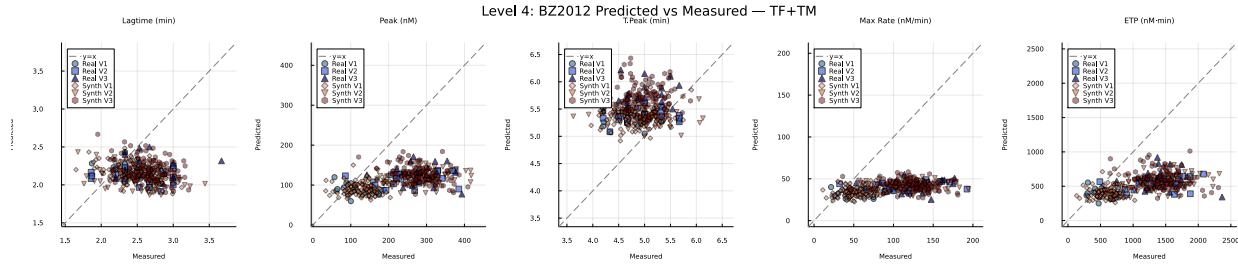


Figure S4: **Level 4: Mechanistic model predicted vs. measured TGA features (TF+TM condition).** Same layout as Fig. 6 in the main text. The model systematically underpredicts peak and maximum rate ($\approx 0.5 \times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. The shared bias indicates that synthetic patients are processed identically to real patients by the mechanistic model.

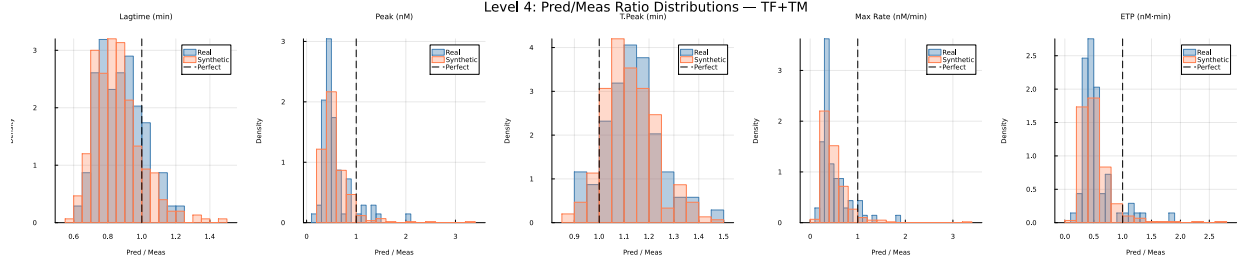


Figure S5: **Level 4: Predicted/measured ratio distributions (TF+TM condition)**. Same layout as Fig. 7 in the main text. Despite the systematic bias in peak and maximum rate, the ratio distributions for real and synthetic patients overlap substantially (88–91% cloud overlap).

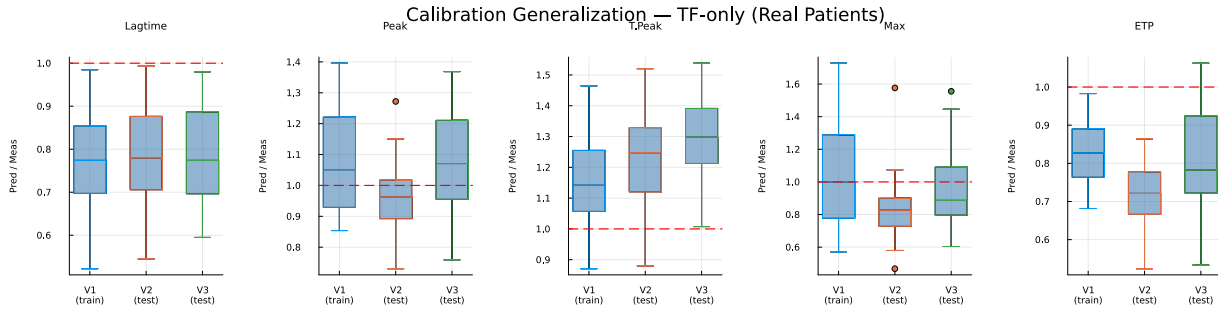


Figure S6: **Level 4: Cross-visit calibration generalization (TF-only)**. Boxplots of the predicted/measured ratio by visit for real patients. The model was calibrated on Visit 1 (“train”) only. Ratios near 1.0 at Visits 2 and 3 indicate that the population-level calibration generalizes across pregnancy timepoints.

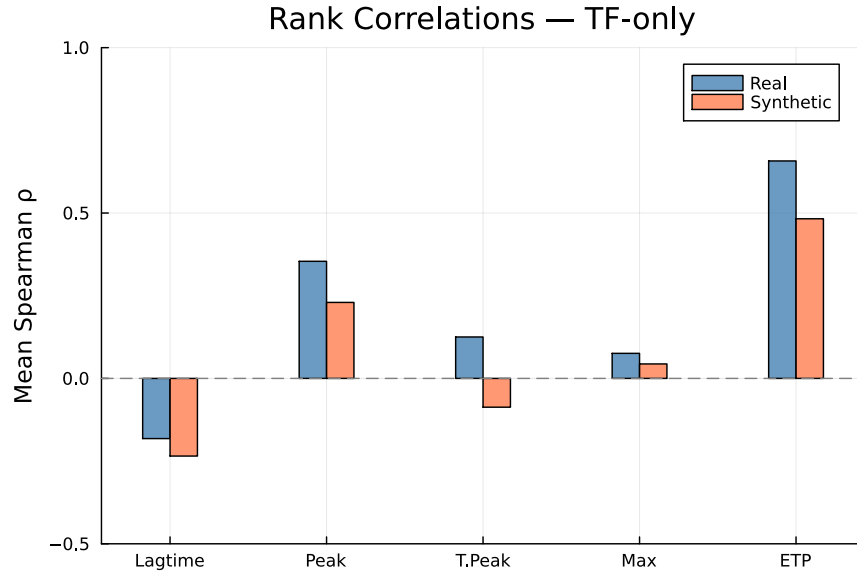


Figure S7: **Level 4: Spearman rank correlations (TF-only)**. Mean Spearman ρ between predicted and measured TGA features, averaged across visits, for real and synthetic patients. ETP is the best-predicted feature ($\rho \approx 0.6-0.8$). Weak rank correlations for other features are expected given the population-level (not patient-level) calibration.

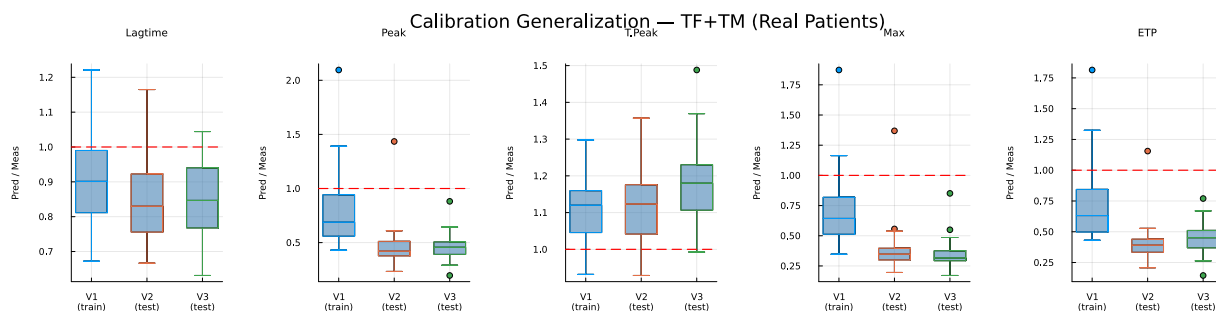


Figure S8: **Level 4: Cross-visit calibration generalization (TF+TM)**. Same layout as Supplementary Fig. S6. The TF+TM calibration does not generalize as well as TF-only, particularly for peak and maximum rate, due to overcorrection of the protein C pathway.

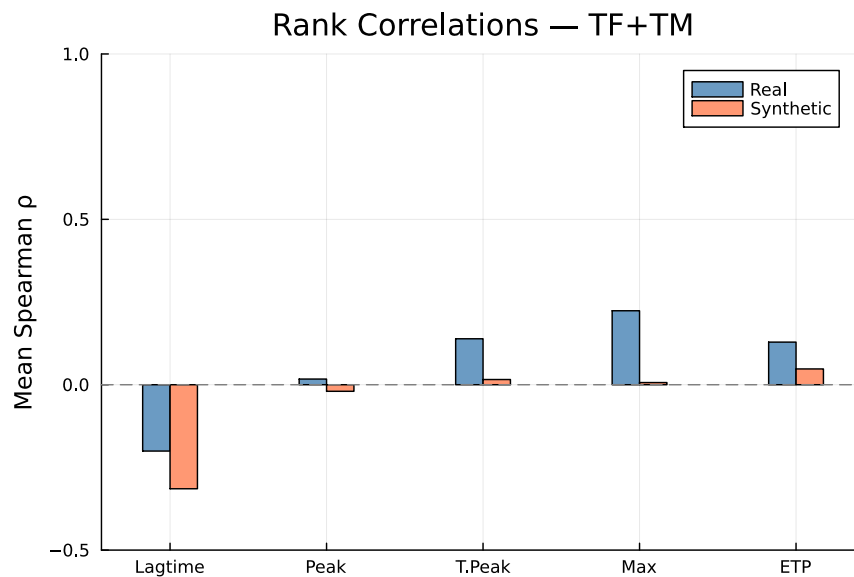


Figure S9: **Level 4: Spearman rank correlations (TF+TM)**. Same layout as Supplementary Fig. S7. Rank correlations under TF+TM are generally weaker than TF-only, consistent with the poorer calibration generalization under this condition.

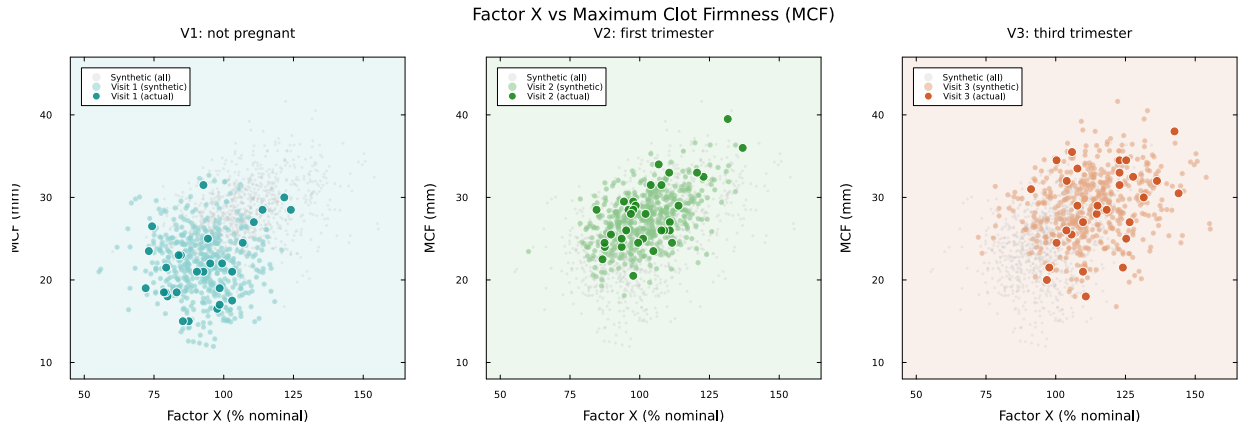


Figure S10: **Factor X vs. Maximum Clot Firmness (MCF) trajectory.** The characteristic pregnancy-driven shift in the Factor X–MCF plane is preserved in SA-generated synthetic patients, demonstrating that biologically meaningful multivariate trajectories are captured by the generation pipeline.

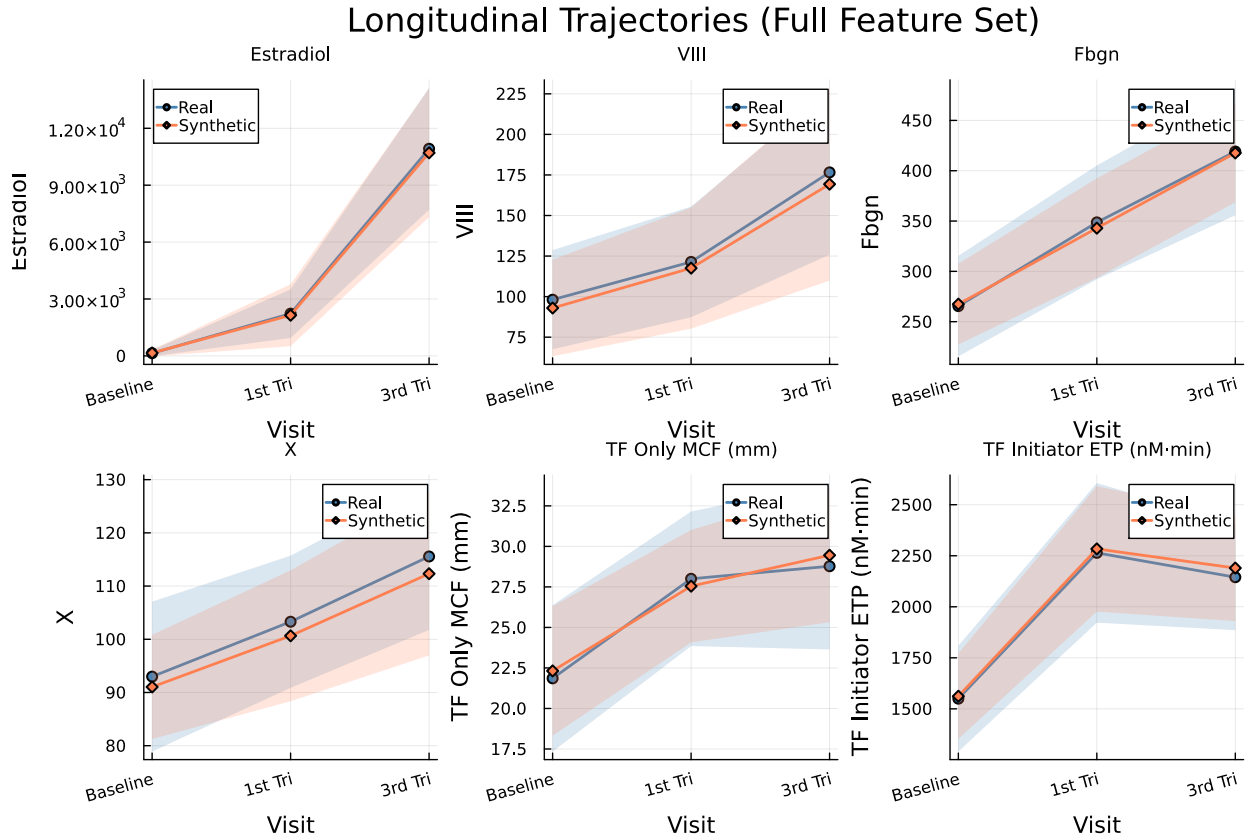


Figure S11: **Full longitudinal trajectories.** Individual patient trajectories (all 72 features) across three visits for both real and SA-generated synthetic patients, demonstrating coherent longitudinal patterns.